

Homework 8

Instructions

Complete the following problems and submit to Gradescope your work as a file named `hw8.Rmd`.

A file with the wrong name or a bug will prevent the autograder from being able to run your submission. Check that your file name is correct, and check for bugs clearing your memory and then running your file from start to finish. It's also a good idea to check that your file "knits."

Note: Do not use any loops and hard-code as little as possible.

Note 2: Before each subquestion, set the random seed to 42. This will affect autograding!

Problem 1

The file `american_delays.csv` contains the lengths of arrival delay (in minutes) of American Airline flights arriving into Dulles International Airport on the Friday, Saturday, and Sunday following Thanksgiving in 2022.

a

Determine the 90% bootstrap confidence interval for the interquartile range of the lengths of arrival delays using these 2022 holiday travel days using 10,000 repetitions.

- call your 12×10000 matrix of resamples `boot_samp`.
- call your sample IQR `samp_IQR`, and finally,
- call your bootstrap confidence interval `boot_IQR_ci`

b

What do you suggest doing with the negative endpoint of the interval? If you think you should scootch it up to 0, assign `TRUE` to `gottaScootch` Otherwise assign to it `FALSE`

Problem 2

The file `southwest_delays.csv` contains the lengths of arrival delay (in minutes) of Southwest Airline flights arriving into Dulles International Airport on the Friday, Saturday, and Sunday following Thanksgiving in 2022.

a

Using these holiday travel days values, is there evidence that the 80th percentile of arrival delay lengths is a late arrival using 10,000 repetitions? Use a significance level of .05.

- Call your 16×10000 matrix of resamples `boot_samp2`;
- call your sample quantile `samp_80th`, and finally,
- call your p value `pval`

b

Is your conclusion in part a indisputable? Explain in a markdown in your `.Rmd` file.

Problem 3

Refer to the American and Southwest arrival delays into Dulles on the days after Thanksgiving 2022 data from the previous problems. Conduct the appropriate randomization test to determine if there is evidence that the 90th percentile of arrival delay lengths is different for the two airlines. Use a significance level of .05.

- Call your simulations from the null `null_sims`;
- call your test statistic `test.stat`, and finally,
- call your p value `pval2`

The autograder is assuming you are setting your seed to 42, so the test will be stricter than you're accustomed to. In particular, the order in comparing two groups will matter. Always use southwest before american, and you should be okay.